



THE COPPERBELT UNIVERSITY
SCHOOL OF GRADUATE STUDIES

CS 594: CLIMATE MODELLING AND DATA ANALYSIS

Practical Exercise 5: Statistical Downscaling of Climate Projections

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Statistical Downscaling of Climate Projections for Mount Makulu, Zambia

1. Introduction

Climate-model projections provide important information for understanding how temperature and precipitation may change in the future. However, regional climate-model outputs can contain systematic biases relative to observations at specific locations. Statistical bias correction is therefore useful for improving the representation of local climate conditions before interpreting future projections.

This practical applied Empirical Quantile Mapping (EQM) to daily maximum temperature (Tmax), minimum temperature (Tmin), and precipitation for Mount Makulu, Zambia. The analysis used six General Circulation Models (GCMs) dynamically downscaled with the SMHI-RCA4 Regional Climate Model. EQM was calibrated against observed station data for the 1981–2010 baseline period and subsequently applied to projections for 2041–2070 under RCP4.5 and RCP8.5. This follows the practical requirement to calibrate EQM using the historical baseline and apply the fitted monthly transfer functions to future projections.

The main purpose of the analysis was to evaluate projected mid-century changes in temperature and precipitation and assess agreement among the climate models.

2. Methodology

CORDEX daily climate data were extracted from the grid point nearest to Mount Makulu (15.55°S, 28.25°E). The nearest model grid point was approximately 15.40°S, 28.16°E. We converted daily Tmax and Tmin from Kelvin to degrees Celsius, and precipitation from $\text{kg m}^{-2} \text{s}^{-1}$ to mm/day. Six GCMs were analysed: CanESM2, IPSL-CM5A-MR, MIROC5, MPI-ESM-LR, NorESM1-M, and GFDL-ESM2M. We bias-corrected each model separately before calculating the multi-model ensemble.

Monthly EQM transfer functions were calibrated using the 1981–2010 period. We corrected temperature values by mapping the modelled empirical distribution onto the observed distribution. Precipitation correction additionally considered wet-day occurrence using a 0.1 mm/day threshold. The fitted corrections were then applied to daily projections for 2041–2070.

Comparison of the raw, bias-corrected and observed climatologies showed that EQM substantially reduced historical model biases, particularly for Tmax and Tmin.

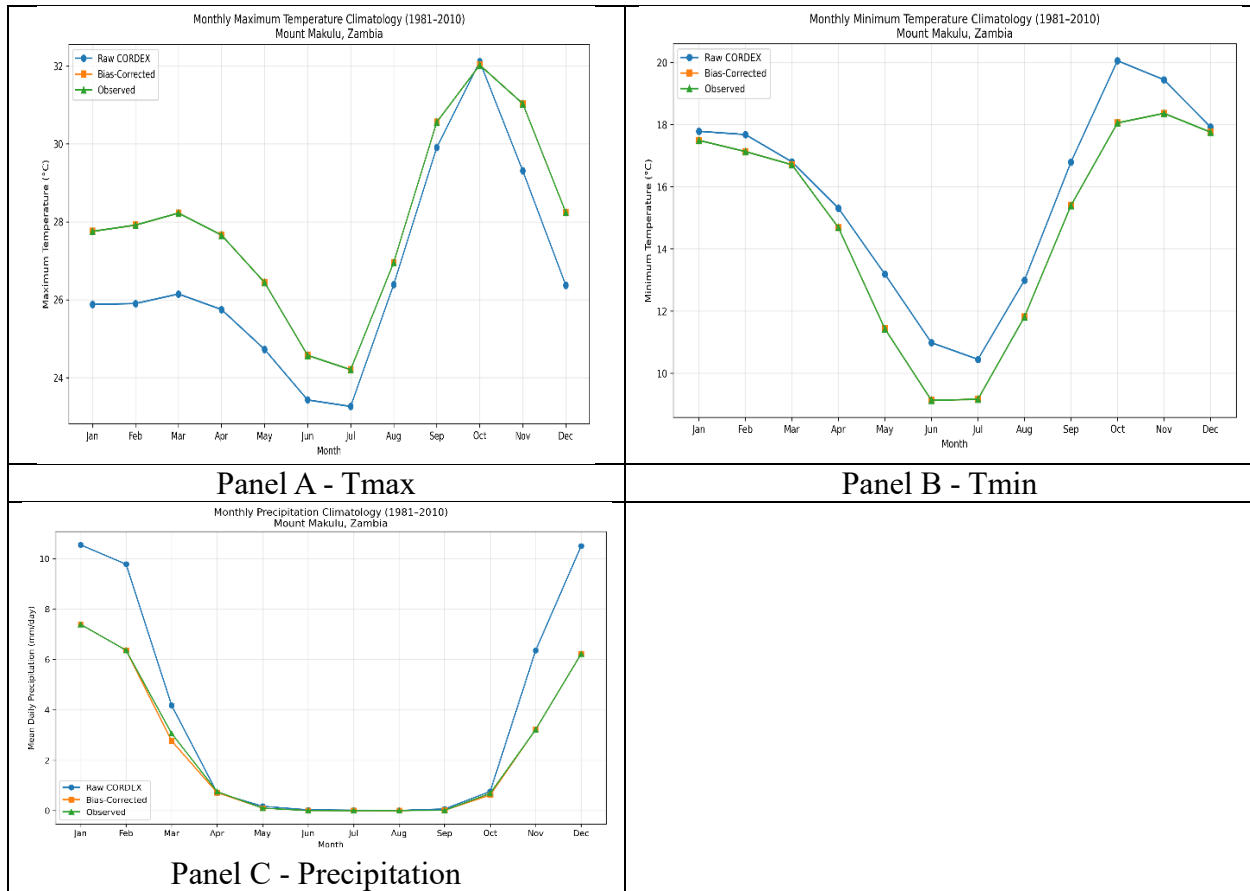


Figure 1. Monthly raw, EQM bias-corrected and observed climatology for Mount Makulu during 1981–2010.

3. Mid-Century Climate Projections

3.1 Temperature Changes

The projections indicate a clear warming signal at Mount Makulu by 2041–2070. The observed mean Tmax for 1981–2010 was 27.95°C. Under RCP4.5, the multi-model mean Tmax increased to 29.61°C, corresponding to approximately 1.66 °C of warming. Under RCP8.5, mean Tmax increased to 30.39°C, representing warming of approximately 2.44°C.

Minimum temperature showed a similar response. Mean observed Tmin was 14.77°C, increasing to 16.51°C under RCP4.5 and 17.21°C under RCP8.5. These values correspond to increases of approximately 1.74°C and 2.44°C, respectively.

Variable	Baseline 1981–2010	RCP4.5 2041–2070	RCP8.5 2041–2070
Mean Tmax (°C)	27.95	29.61 (+1.66)	30.39 (+2.44)
Mean Tmin (°C)	14.77	16.51 (+1.74)	17.21 (+2.44)

Annual precipitation (mm)	840.35	901.65 (+7.30%)	914.85 (+8.87%)
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The monthly projections indicate that warming occurs throughout the annual cycle rather than being restricted to a particular season. The highest projected Tmax occurs around October, reaching an ensemble monthly mean of approximately 34.48°C under RCP4.5 and 35.74°C under RCP8.5. RCP8.5 generally produces greater warming than RCP4.5.

The six individual models differ somewhat in the magnitude of projected warming, but their results show a consistent increase in future temperature. Under RCP4.5, model-average Tmax ranges from approximately 29.28°C to 29.95°C, compared with 29.91°C to 30.80°C under RCP8.5. This relatively limited spread indicates stronger agreement among the models regarding the temperature response.

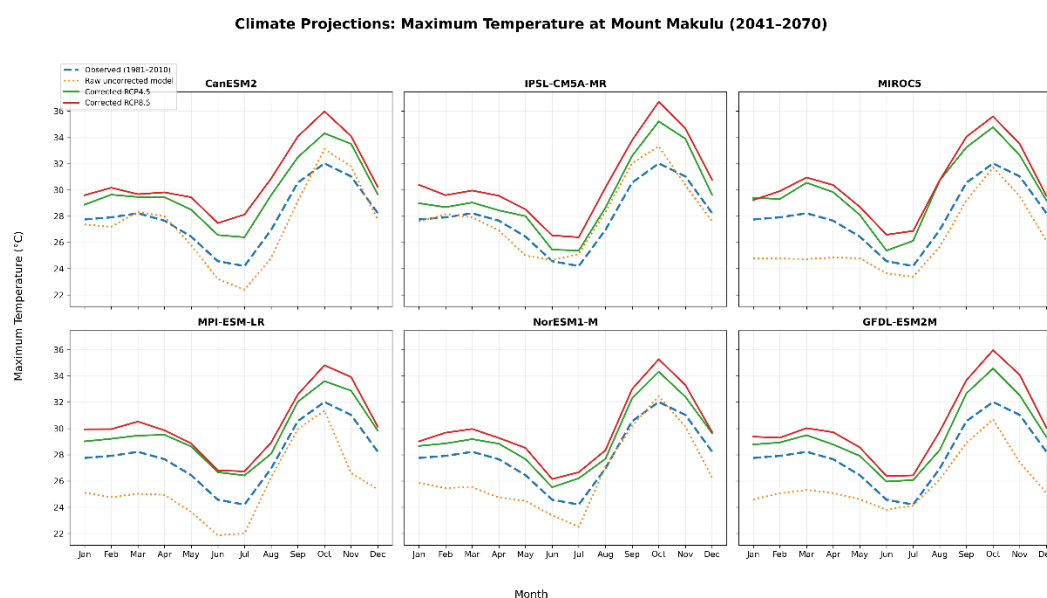


Figure 2. Bias-corrected maximum-temperature projections from six GCMs for Mount Makulu during 2041–2070 under RCP4.5 and RCP8.5.

3.2 Precipitation Anomalies

Projected precipitation changes are less pronounced than temperature changes. Mean annual observed precipitation during the baseline period was approximately 840.35 mm/year. The multi-model ensemble projects annual precipitation of approximately 901.65 mm/year under RCP4.5 and 914.85 mm/year under RCP8.5.

Relative to the baseline, these values represent precipitation anomalies of approximately +7.30% under RCP4.5 and +8.87% under RCP8.5. The ensemble therefore suggests a modest increase in annual precipitation by mid-century.

The seasonal distribution of precipitation does not change substantially. Most rainfall remains concentrated during the wet season, particularly from approximately November to March, while

June–August remain very dry. The projected changes therefore suggest changes in rainfall amounts rather than a major alteration of the existing seasonal cycle.

However, precipitation projections show considerably greater inter-model variability than temperature. Under RCP4.5, individual model annual precipitation ranges from approximately 824.49 to 1002.32 mm/year, while RCP8.5 projections range from approximately 871.93 to 1007.04 mm/year. Therefore, the ensemble-average increase should be interpreted cautiously.

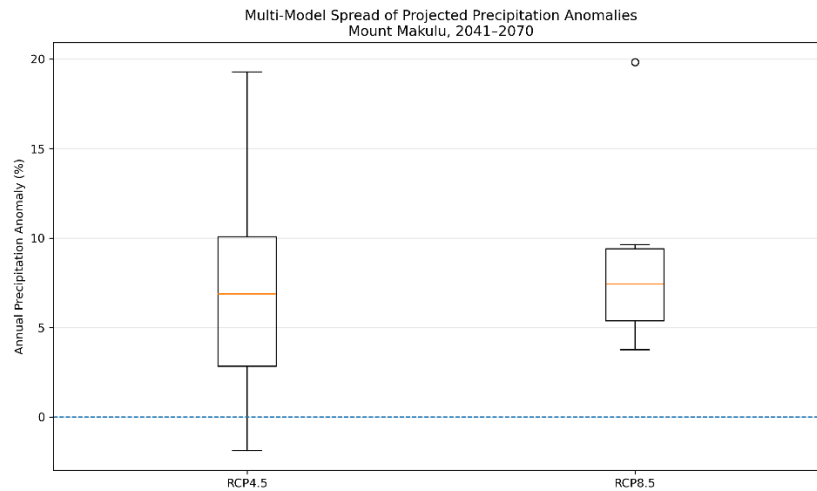


Figure 3. Multi-model spread of projected annual precipitation anomalies for Mount Makulu during 2041–2070.

4. Discussion

The results indicate that temperature change is the clearest mid-century climate signal at Mount Makulu. Both Tmax and Tmin increase under RCP4.5 and RCP8.5, with larger increases under RCP8.5. Importantly, the warming signal is present across the six GCMs, despite differences in the magnitude of the projected change.

Precipitation presents a less certain response. Although the multi-model ensemble indicates increases of approximately 7–9%, the spread among individual models is considerably greater than for temperature. Consequently, the projected precipitation increases should be interpreted as an ensemble tendency rather than an exact prediction of future rainfall.

The analysis also demonstrates the importance of bias correction. Comparison with observations showed that EQM substantially reduced historical biases before the model simulations were used to assess future conditions. However, EQM is a statistical correction and assumes that relationships between historical model biases and observations remain applicable in the future.

Additional uncertainty arises from missing observations in the baseline station dataset and from using a single RCM, SMHI-RCA4. The six-GCM ensemble captures differences among driving

global models but does not represent the full range of uncertainty that could arise from using different regional climate models or alternative bias-correction methods.

5. Conclusion

The statistical downscaling analysis indicates a clear warming of Mount Makulu by 2041–2070. Relative to the 1981–2010 baseline, mean Tmax is projected to increase by approximately 1.66°C under RCP4.5 and 2.44°C under RCP8.5, while Tmin is projected to increase by approximately 1.74°C and 2.44°C, respectively.

Annual precipitation shows a smaller and less certain response. The multi-model ensemble indicates increases of approximately 7.30% under RCP4.5 and 8.87% under RCP8.5, while individual models show substantial variability in projected rainfall.

Overall, the results suggest that warming is more consistent and robust across the six climate models than changes in precipitation. The analysis therefore highlights both the usefulness of EQM for producing locally relevant climate projections and the importance of using a multi-model ensemble when interpreting future climate uncertainty at Mount Makulu.